



Integrating SCM with Internet of Things: Implication on HR Management

Preeti Khanna
School of Business Management, NMIMS
India
preeti.khanna@nmims.edu

Ashu Sharma
School of Business Management, NMIMS
India
ashusharma74@gmail.com

ABSTRACT

Business imperatives are changing for every industry across the globe. The focus is on transforming supply chain by improving its performance and efficiency. In today's dynamic world Internet of Things (IoT) is significantly contributing by the use of connected devices. IoT facilitates with real time information using automated and intelligent seamless processes in the supply chain. The objective of this study is to understand the benefits and challenges of integrating IoT in supply chain. The research has done an extensive literature review to explore the role of IoT as an enabler for end to end supply chain. Technology cannot be functional without the intervention of channel partners. Therefore, the paper tries to explore and develop an integrated framework to link IoT with SCM and further attempts to see its implication on HR Management. Limited research has assessed the inter linkages between SCM performance and HR Management because of the intervention of technology, especially IoT. The researcher's anticipate that finding of this paper will be practical and will be beneficial for researchers, academicians as well as practitioners. The research would certainly help practicing managers to comprehend the significance and role of IoT in gaining SCM effectiveness by effective HR Management. The study would also broaden the understanding of interdisciplinary approach at the academic level. This transformation can be driven by the perception of motivated individuals and channel partners in end to end supply chain. Satisfied HR at individual and channel level is ultimate and vital goal of every supply chain.

CCS Concepts

Supply Chain Management → Internet of Things → Role of HR Management → Influencers of IoT → Developing a Framework → Best Practices for Supply Chain Performance.

Keywords

Supply Chain Management (SCM); Internet of Things (IoT); HR Management; Integration; Collaboration; Best Practices.

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1. INTRODUCTION

In the current competitive scenario supply chain management assumes a significant importance and calls for serious research attention, as organizations are challenged with finding ways to meet ever-rising customer expectations at a manageable cost. With rise of technology innovations, organizations adopt numerous business improvement methodologies to improve the business performance. In today's digital well-connected world, technology is driving the change in global business strategies. New trends of adopting Internet of Things (IoT) is transforming the entire business, here the focus is on transforming the Supply Chain Management (SCM) using IoT. Beforehand makers were the drivers of the supply chain network - dealing with the pace at which items were produced and disseminated. Today, customers are very demanding, and makers are scrambling to meet customers' requests for choices/styles/highlights, speedy demand satisfaction, and quick response in delivery alongside the ongoing follow up and checking. IoT in supply chain for collaborating and coordinating with the suppliers, manufacturer and the customers is the need for the day for every supply chain. Alignment of supply chain strategy with the business strategy is necessary for improved sustainability of supply chain.

Winning supply chain know that in the advanced age know achievement originates from more than simply finishing an agenda of technological innovation. It depends on individuals. In a disruptive world, workforces must be motivated to adjust, to grasp new business methodologies and constantly learn new skillsets [1].

The aim of this research paper is to study the implication on HR management by integrating IoT practices in SCM. In the present global scenario, it is difficult to sustain the success of SCM without technology intervention in the real time. But the IoT based supply chain can facilitate and also pose challenges to the managers. One of the objective of this paper is to discuss the positives and describe the challenges faced by the channel partners in this journey and what must be done to overcome these challenges if supply chain needs to absorb IoT efficiently. Let's first begin with the background of SCM and IoT in detail.

1.1 Supply Chain Management (SCM)

SCM has got consideration since the early 1980s, yet adroitly the administration of supply chains is not especially looked into, and numerous authors have highlighted the need of clear definitional constructs and reasonable frameworks on SCM ([2], [3], [4], [5], and [6]). Adoption of supply chain administration rehearses in

commercial enterprises has relentlessly expanded subsequent to the 1980s. Various definitions are proposed and the idea is talked about from numerous points of view. However, different analysts ([7], [8], and [9]) gave point by point review on supply chain management writing. These papers characterize the idea, principals, nature, and improvement of SCM and demonstrate that there is a serious exploration being directed far and wide in this field they fundamentally researched advancements in the theory and application of supply management.

SCM is defined as a set of activities that span enterprise functions from the ordering and receipt of raw materials, manufacturing of products, through to the distribution and delivery to the customer. These activities are associated with the material flow and information flow [10]. Originally, it is a set of synchronized activities for integrating suppliers, manufacturers, transporters, and customers efficiently so that the right product or service is delivered at the right quantities, at the right time, to the right places.

Other Researchers ([11], [12], and [13]) view SCM as a collaborative and cooperative methodology, spreading over crosswise over firm limits, incorporating different channel partners in the supply network, which altogether add to enhanced item quality, shorter lead times and a higher responsiveness of the supply chain, at lower cost and expanded consumer loyalty levels.

Research demonstrates that creative advancements ought to be used to enhance contemporary techniques for supply chain. Research also shows that SCM would not be conceivable without advances in data frameworks and innovation [14]. The ultimate objective of SCM is to achieve sustainable competitive advantage. Hence today, SCM is increasingly seen to be a strategic, highly integrative management area that exceeds any single functional perspective. Therefore, technology keeps on enhancing imperative in SCM as a proficient supply chain which is portrayed by constant coordinated effort and integration ([15], [16], and [17]).

1.2 Internet of Things (IoT)

Internet of Things (IoT) was devised to connect the things in the physical world to the Internet via sensors that could gathered and then represent as and when required. That was in 1999, when Kevin Ashton [18], co-founder of the Auto-ID Center at MIT, proposed the term “internet of things”. Today, because of this characteristic IoT is reshaping and redesigning the global supply chain in almost every sector of the economy. There is huge evidence that the opportunities and applications for the IoT are arising all around the industry including retail to manufacturing to hospitality. And there is a huge potential and feeling that IoT is the future for supply chain. It has come out very clearly that smart and intelligence devices in supply chain can make critical decisions and adjustments in response to complex scenarios, and this in turn enables channel partners to improve further within the network. These devices have the ability for continuous monitoring that can sense constraints, if any in the respective physical network and produce intelligent response and alarms to the respective stakeholder.

IoT embedded in products can send real time customer feedback to manufacturers on the usage and success / failure of a product. This can aid manufacturers in design decisions made in the early cycles of product development to enhance and optimize for supply and manufacturability. In addition, this information can be shared

with key suppliers during product innovation and development processes to further improve product capabilities.

As the focus of this paper is to discuss about the integration of SCM with IoT, detail literature were studied. According to a recent Deloitte IoT research paper [19], “Modern SCM can be not only about getting products faster, cheaper, and of better quality but also about getting managers the right information at the right time, so that they can better make informed supply chain decisions”. In this regard, IoT is being used for managing, monitoring and improving decision making for businesses by transforming how people interact with their surroundings (including devices, machines, etc.). As mentioned in one of the report that IoT is transforming various activities and task across all the sectors and improving productivity, efficiency, and profitability [20].

IDC expectations is that technology and services revenue from IoT will expand from US\$2.3 trillion from 2014 to US\$4.6 trillion by 2018 at a 19.0% compound annual growth rate (CAGR) globally. According to researchers, IoT is estimated to reach 50 billion connected devices by 2020 [21] and the potential economic impact of IoT technologies will be between \$2.7 to 6.2 trillion per year by 2025 [22]. This explains that there is huge opportunities for consumers, as well as for the developers. IoT is used in many spheres of daily lives as well as from businesses point of view. Some domains are discussed below:

- Connected the Production Floor to enhance the effectiveness: IoT enables real time monitoring to understand what is happening at a given time, at a given location, or at a given floor. For example, the performance of machines, ambient or extreme conditions, energy consumption, status of inventory, or the flow of materials, goods, throughput time, communication with workstations, problem detection, could be detected and alarmed to the concerned one. Sensors can alert managers or supervisor or labor with machines on shop floor that a physical asset in the factory is exceeding or not following acceptable norms (may be of temperature or humidity or power consumption or is malfunctioning). This has major implications in terms of collaborative planning forecasting and replenishment (CPFR), a key metric of productivity in supply chain.
- Monitoring: Monitoring is the essential key in supply chain being goods, or information. Today this is possible via the intelligent network empowered by sensors that can be used to notify to the essential stakeholders immediately and make the necessary adjustments to avoid a line down situation. Cisco has already started a pilot program in its Malaysia site, beginning this May, to see how the IoT network will benefit them [23].
- Handling Logistics Services Effectively: By connecting through IoT the different players and many moving parts within the logistics ecosystem can make network more flexible and intelligent. This enhances further supply chain professionals to monitor important details as and when required. Transportation Management System will help in communicating real time information with Logistics Providers like 3PL, 4PL, etc.

for optimizing cycle time, visibility, Full truck load (FTL) shipments.

- *Physical Security:* IoT enables end users to lock and unlock doors (on shop floor, or offices) from anywhere using a mobile app. Physical security which is the biggest challenge today, with the help of IoT these kinds of intrusion detection serve as a smart alarms (sensors attached to a child's clothing or toy or wall or a pet's collar, for example, can be configured to alert the user if they have strayed from the premises [24]). The August Smart Lock, [25] is mentioned as a popular consumer application of IoT which allows the consumers to provide access to their home or private area only to trusted parties remotely. This could also provide new delivery options in last-mile logistics. There can be more transparency and visibility of delivery made at the buyer end using hand held devices integrated with IoT.

A number of leading global manufacturers—including the likes of Bosch, Cisco, FCA (Fiat Chrysler Automobiles), GE, General Mills, Harley-Davidson and Siemens—are early adopters of smart manufacturing in their plants [26].

The IoT is not very new, but how it is being integrated in the supply chain to gain competitive advantage and bringing with it a level of efficiency that supply chain professionals from years past have not experienced is new beginning. Most of the time much discussion on IoT is conducted at a high level of abstraction or in a very generic context. However, it is as important to discuss the people aspect while using IoT in SCM. IoT as an enabler in supply chain (which will be addressed in the later part of the paper) will depend on the attitude, skill set, knowledge, usage, etc. of channel partners in the chain. What might hold true for IoT systems and technologies in the inbound process may not be necessarily be true for IoT systems in outbound process or in home automation or in vehicles and public transport systems. HR management concerns may change as per the requirements / expectations while working in cross functional team.

This paper aims to study and understand the implications on HR management while transforming SCM by integrating with IoT. Section 1, above covered conceptual understanding of SCM and IoT. Next Section, i.e. section 2 discusses the detail literature review in the field of integrated SCM, and IoT. Also this section try to project the related and existing work in the area of linking SCM with HR management. After understanding the research gap, section 3 list the research objectives and methodology of the study. The paper discusses analysis in detail for each of the objectives (section 4) and then (section 5 and 6) concludes by summarizing the findings attained and also considers the future scope and limitations. Section 4, analysis and finding includes the following four objectives i.e. (i) description of a 3-layered architecture of IoT along with its benefits across various functions in supply chain, (ii) positive and negative influencing factors of IoT while integrating with supply chain, (iii) proposing a framework for understanding the implication on HR management to use IoT in supply chain, and (iv) proposing the best practices for motivating individuals and the channel partners to use IoT in supply chain. To the best of our knowledge, the study is novel which connects and discusses the implications on HR management while transforming SCM using IoT.

2. LITERATURE REVIEW

2.1 Integrating SCM with IoT

SCM rotates around effective mix of suppliers, producers, distribution centers and stores. The challenge in supply chain coordination is to co-ordinate activities over the supply chain so that the business can enhance execution: lessen costs, improved service level, diminish the bullwhip impact, better use of assets and successfully react to changes in the commercial scenario [27]. Supply chain coordination is depicted as 1) supply chain individuals utilizing techniques empowering them to cooperate to improve their aggregate execution in creation, dissemination and support of the final product [28]; 2) the consistent stream of items and data from supplier to customer [29]; and 3) coordination systems, all of which infers that business procedures ought to be streamlined and interconnected, both inside and outside the organization's limits [30]. The premise of integration can in this way be portrayed by co-operation, coordinated effort, data sharing, trust, associations, shared innovation and a principal move far from overseeing individual utilitarian procedures to overseeing coordinated chains of processes [31]. Successful supply chain coordination happens when the players understand that SCM must turn out to be a piece of all the business' strategic planning processes, in which the goals and policies are together decided in view of the end customers' needs and what the supply chain overall does well [32].

Research uncovers that data sharing can generously enhance supply chain performance [33]. Researchers additionally conceded to the significance of information sharing for accomplishing supply chain integration, increasing the coordination among channel partners in the supply network, and prompting a superior performance [34]. As per Liu et al., [35] the real success factor for a supply network incorporate broad information sharing capabilities and advanced inter and intra channel systems to empower better information sharing.

As the focus area of the study is SCM, hence the literature focuses around an IoT based SCM. Some recently conducted reviews about IoT (e.g., [36], [37], [38], [39], [40], and [41]), they focus on high level open issues. Few researchers try to connect IoT with SCM. According to L'opez et al. [42] smart technology can be utilized to upgrade supply chains by enhancing nature of information and perceivability of products, and expanding precision and opportuneness of data and information about the business procedure. Haller et al. [43] state that extra information supplied by these smart items, empowers an organization to better comprehend what happens in 'this present reality'. Because of which supply chain turn out to be more transparent. Jedermann, R. [44] talk about that IoT can disentangle reviewing and examining of supply chain and can permit smart items a level of opportunity in decision making. Atzori et al. [45] clarify how the Internet of Things can be connected in supply chains; Haller et al. [43] clarify the impact of real world visibility on supply chains and the impact of business procedure decomposition on business procedures. Wagenaar, J. [46] talks about the advantages the IoT has on supply network and looks how these advantages can build revenue inside a supply chain.

IoT- enabled warehouses and supply chains are still in its early stages. Therefore, the industry is still in the process of trying out different applications and approaches in an effort to figure out what makes the most sense. The global logistics market is forecast

to grow at a CAGR of 8.4% over 2015-2019 (Source: Technavio, [47]). IoT is set to boost supply and logistics operations by US\$1.9 trillion over the next decade (Source: DHL and Cisco Consulting Services, [48]). Each coin has two sides, there are concerns too which has needs to be equally addressed. The security of the IoT is an issue that's only going to get more and more prominent, as an increasing number of connected devices enter the market. Also, another threat identified is the lack of architecture standards for the connectivity in the IoT [49].

As per the two schools of thoughts whether technology or people, which one make a significant impact on supply chain, it describes further that none independently can make the supply chain function. It is people as well as technology both side have to drive the supply chain towards attainment of its objectives. Thus, it is important to preserve all channel partners trust and loyalty in order to succeed in the holistic supply chain. They must appreciate and understand the power of the new technology and how this will take their services a step further. Few researchers tried to understand how HR management is linked to supply chain in order to enhance the collaboration and performance further.

2.2 Integrating SCM with HR Management

Literature discusses about how HR management could enhance the working condition of SCM. Researchers like [50], [51], [52], and [53] were among the first one to link HR Management to SCM. Using case studies, these scholars suggested that the development of SCM posed great challenges for HR and their practices contributed to building and facilitating supply chain relationships and collaboration. Cohen et al. [54] found that the right incentives could lead to desired supply chain behavior. Jin et al. [55] suggested that human capital could be leveraged to improve manufacturers' flexibility in supply chains. Barnes and Liao [56] and Lorentz et al. [57] considered competency and skill as enablers of SCM. Thornton et al. [58] identified five types of employee work behavior that impeded the success of supply chain relationships. Fu et al.'s [59] empirical study found that HR Management could improve supply chain performance through team formation and implementation.

The aim of supply chain integration is to achieve highly coordinated processes which can provide maximum value to customers effectively and efficiently [60]. According to Chen et al. ([61] and [62]) supply chain relationships exist both internally and externally which includes both internal and external integration. Internal integration means "the degree to which a firm can structure its organizational practices, procedures and behaviors into collaborative, synchronized and manageable processes in order to fulfill customer requirements" [63]. Internal integration mainly involves information sharing, cross-functional cooperation for improved processes or to develop new products [63]. External integration includes customer and supplier integration. It refers to "the degree to which a manufacturer partners with its external partners to structure inter-organizational strategies, practices and processes into collaborative, synchronized processes" [64]. Zhao et al. [63] observed that data sharing, synchronized cooperating with customers and suppliers to together resolve issues and encourage operations were critical issues of external integration. However, working with customers differs from working with suppliers in terms of nature and focus.

These studies provided valid evidence that HR Management played a significant role in smooth functioning of SCM. This dynamic environment requires effective communication, team management, and constant lifecycle innovation. Human factors insights in these areas are critical for the effective development of supply chain. The importance of technology intervention along with the right skilled people has increased significantly in the entire supply chain. Based on review, research questions have been framed and listed in the next section.

3. RESEARCH QUESTIONS AND METHODOLOGY

The aim of the research paper is to understand the implications on HR management faced in transforming SCM using IoT. The scope of this study includes following research objectives which are as follows –

Objective 1: To understand broadly a 3 - layer architecture of IoT along with its benefits across various functions in supply chain

Objective 2: To list the positive and negative influencing factors of IoT while integrating with supply chain.

Objective 3: To develop a framework for understanding the implication on HR management to use IoT in supply chain.

Objective 4: To frame the best practices for motivating individuals and the channel partners to use IoT in supply chain.

To meet the above mentioned objectives, secondary data was collected from diverse sources like case studies, press releases, research papers, white papers, web sites, blogs, journals, etc. related to IoT, SCM and HR Management. An in-depth comprehensive study was done to identify the linkages between various factors for HR Management which enhances the usage of IoT in SCM.

The research is addressing a contemporary topic and is about people and for people (with reference to IoT in SCM) which is not heavily discussed and empirically tested in literature yet. The plan of study concentrates on the following three phases:

Phase 1: The existing literature, company reports, websites, white papers and the organization practices were seen and studied closely. These were drilled in depth and analyzed further to understand what is happening, and how SCM is being transformed by the intervention of technologies. As mentioned in the report of Accenture technology vision 2016 [1], "Intelligent automation will enable enterprises to innovate and evolve by increasing their agility, reducing the complexity of systems and operations, accelerating their time to market, and creating the ability to experiment continually with new products and services". Similar review reports suggest that technology does transform supply chain. Review of the related concepts of SCM, IoT and HR Management were seen in the industry.

Phase 2: The broader view of IoT architecture with its elements were understood. The two level mapping i.e. IoT with SCM were framed. Then, the complete framework i.e. (IoT in SCM) was mapped with the third level which is HR Management to know how people perceive these new trends. The HR Management (i.e. channel partners) perception of accepting this change was discussed. Finally, an integrated framework which links SCM, IoT and HR Management was proposed.

Phase 3: Based on the existing practices, researcher views and findings, suggestions were made to HR Management, 'how to handle IoT devices', addressing the pros and cons of these in integrated SCM.

4. ANALYSIS AND FINDINGS

The study aims to address the research questions raised in section 3. Next, the paper discusses each of the objectives and its analysis in detail.

Objective 1: To understand broadly a 3 - layer architecture of IoT along with its benefits across various functions in supply chain.

As seen from the literature as well as from the Industry practices across the globe, now a day's supply chain management deals with huge data and digital trends are affecting them. IoT has been one of the trends which is in rising demand. As mentioned in one of the article by Michael Weyrich and Christof Ebert, [49] IoT will boost further a tremendous amount of innovation, efficiency, and better productivity by seamlessly connecting devices. The architecture of IoT suggested to be open enough so as to use open protocols to support a variety of existing network applications. The three main layer of IoT considered to be more useful in SCM context as Sensing layer, Network layer and Application layer having specific functionality. These three layers are as follows: -

- **Application Layer:** It is the top most layer, providing a variety of IoT applications such as monitoring the status of objects or products, operational control, manager's enquiries, finding the status of product, location details, and other value added applications. In SCM it can effectively handle - SMART logistics, Smart Transportation, Inventory Management, Communication Technologies, etc.
- **Network Layer:** It is located above the sensing layer and below application layer. It is the main body which include its physical network, system software, open communication protocols and sophisticated technical mechanisms. Internet layer also acts as a transmission carrier. It is used to exchange the information between things / connected devices and things / connected devices, things and applications and even applications and applications. It make use of Mobile Network, Satellite Network, Internet, VPN, etc.
- **Sensing Layer:** It is located at the bottom of the layered architecture. It includes RFID, Intelligent Sensors and Terminals, Networks, Wearable Devices (or types of sensors). These are actually the sensors / devices which is attached in handling inventory, logistics, transportation, etc.

IoT intervention in supply chain could enhance the functionality associated in SCM as shown in Figure 1 (see Appendix). It displays numerous benefits of IoT in most of the supply chain functions. IoT benefits end to end supply chain by optimizing cycle time, lead time, transit time, inventory delays, stock outs, demand distortion, transportation, information, etc. and also integrating the supply chain through better coordination and collaboration between the buyer-supplier link.

Objective 2: To list the positive and negative influencing factors of IoT while integrating with supply chain.

There are many advantages of IoT which can help individuals as well as businesses to keep them updated and well informed in this global world. There won't be any surprise that in the near future, IoT will become an integral part of day to day life. However, there is increasing evidence that the IoT will definitely have an impact on various types of Industry, here the focus is on supply chain. The key questions are: how big will the impact be? Will the impact be positive or negative? Here listing all the majority of positive as well as negative influencing factor while integrating IoT with SCM.

Positive Influencing Factor of Integrating IoT with SCM –

- **Reduced cost of operations:** The cost of availing IoT services is much cheaper as it allows the data to be shared between devices and then translating it into required way as requested by end users. The Supply Chain operates at a reduced cost because of real time data transmission.
- **Smaller staff:** The ability to outsource applications reduces the need for people to manage. The IoT application facilitates supply chain to be more technologically driven.
- **Traceability, Transparency, Reputation, Better Monitoring and Marketing:** With IoT, the entire world is open to the service providers, consumers and others for marketing as monitoring is much more feasible and easy. Visibility of supply level in chain helps in better inventory management.
- **Communication and Visibility:** IoT encourages the communication between all partners in the chain. It has ability to have greater insight into exactly what is happening at any stage and at any time in the supply chain.
- **Automation and control hence efficient Inventory Management:** Without human intervention, the connected devices are able to communicate with each other leading to faster and timely output. Automated planning and execution of inventory management reduces subjectivity and hence reduces surplus and stock-outs scenarios.
- **Consolidated information availability any time any location:** it is obvious that having more information helps making better decisions. Transparency of data and information shared with all channel partners leads to informed decisions.
- **Better Quality of Life:** All the applications of this technology culminate in increased comfort, convenience, and better management, thereby improving the quality of life. This improves efficiency and effectiveness of all channel partners in chain.
- **Monitor loss management:** Pain areas and weak links within the supply chain can be identified before they become a problem that impacts revenue and customer satisfaction.
- **New opportunities:** With IoT, the channel partners can acquire a lot of data about the consumer, enabling more customization and hence micro targeting is feasible for the supply chain.

The use of IoT is numerous in supply chain. Though it offers lot of positives but it rises the concern for the privacy of individuals or groups of people too. In some cases, individuals may not even be aware that they were being tracked or monitored. By keeping

these concerns, here follows the major negative influencing factor of IoT being used in concern with SCM.

Negative Influencing Factor of Integrating IoT with SCM –

- Specificity: Channel partners have a very specific need, might not be able to find the same solution across its distribution in supply chain.
- Dependency: Although the quality of service rendered may be good, network and internet outages are an issue with regards to continuous availability of the IoT especially when channel partner is placed in remote locations.
- Lack of standardization: Currently, there is no international standard of compatibility for monitoring equipment. As such there can be compatibility concerns when data is collated from multiple locations.
- Safety and increased risk: As of now, IoT is very much hackable. Many of these threats have not even been identified specifically.
- Complexity: The IoT is a diverse and complex network. Any failure or bugs in the software or hardware will have serious consequences. Even power failure can cause a lot of inconvenience. A failure with one channel will affect the entire chain. This can lead to serious consequences for the supply chain
- Lesser employment of menial staff: The things around the SCM is now more connected and getting smarter. The unskilled workers and helpers may end up losing their jobs in the effect of automation of daily activities.
- Shortage of expertise: It is tough to get the right skill set if the channel partner is located at remote locations.
- Technology savvy: The channel partners if not comfortable with the use can resist for the technology implementation. One channel resistance will have repercussions for entire supply chain technological failure.
- Data privacy and security concerns: Since the service makes extensive use of the internet to host data and applications alike, the threat of security remains the topmost concern for the supply chain. With all of this IoT data being transmitted, the risk of losing privacy increases.
- Technology takes control of life – People's lives will be increasingly controlled by technology, and will be dependent on it and hence can impact on to supply chain.
- Vulnerable to cyber-attacks - Infrastructure that enables the IoT is very vulnerable to cyber-attacks, especially as it embraces modern communication technologies, such as wireless networking and the Internet. This can be fatal for the supply chain.

By looking at both the sides of the coin, there is a need to handle the concerns and issues raised with care for the overall benefit. Of foremost concern is the need to make sure that this integration with HR Management should have most positive and motivational implications from the organization perspective. Next research objective will answer this concern to prevent the negative effects of using IoT at the individual as well as organization level.

Objective 3: To develop a framework for understanding the implication on HR Management to use IoT in supply chain.

With people, at every level, driving change with new tools, techniques, new skills, and new devices used in day to day businesses, leaders will have a critical role to play. Less emphasis on the research related to HR Management and SCM performance could be seen. Reasoning that improvement of SCM originate from the combination of manufacturing procedure and utilization of technology. It stressed performance measurement of SCM to a great extent on issues like optimum utilization of resources, cost reduction and customer satisfaction. The part of improving internal HR to help SCM practices is still liable to be concentrated on legitimately. Past researches scrutinizes into the best interest of organizations driving these endeavors in advancing its SCM performance.

The literature affirms that logistics excellence can be accomplished by viable human asset. It is noted excellent SCM have a solid accentuation on preparing and re-preparing of the workforce. Human resource development and training are considered as a key need in best supply chain. Change is becoming a routine, and therefore, via training best supply chain can stay competitive and can plan for future opportunities. A different study conducted by Michigan State University emphasized on human resource development considered as 'best in-class' in context of SCM performance [65].

This paper fortified that human asset improvement straightforwardly affects the achievement of SCM patterns and practices. These core capabilities incorporated the workforce capacity and ability to be adaptable in skill sets and roles, enlarge their efficiency, redeployment and boundary spanning duties. Later, a report of Ernst and Young, [66] expressed the human resource development as an essential element in the headway of "total quality," by putting concern on total employee involvement, their skills and training. Employees expected to display effectiveness in their work and to a great degree adaptability to a focused domain.

Integrating IoT with SCM have seen many advantages, and to use them at each level especially at lower level and even in remote locations in the supply chain can make more efficient process and produce better results. Monetary and other rewards to improve motivation is a simple idea, but doing it fairly and effectively can be challenging. There are few perception about IoT from HR perspective which have come out during the research.

PERCEPTIONS 'WHAT CHANNEL PARTNERS THINK WHY TO USE IoT?'

SCENARIO 1: Our workforce won't like having an IoT.

The thought of being monitored and measured can be disconcerting and uncomfortable, which is why many partners anticipate trepidation from their workforce when IoT is being used.

What next: The importance of effectively communicating with the workforce in this journey plays an important role.

SCENARIO 2: If integrating IoT closely with SCM doesn't show immediate results, we didn't make a good decision.

It can take a while to get primary and base users up to speed in day to day activity as now each one has to be tech-savvy. And it

can take even longer to try out all the capabilities and see how—or whether—they should work for supply chain operation.

What next: IoT can work wonders but only if it's being used properly and align with the business. Hence paying attention to the fundamentals is most important and necessary.

PERCEPTION 'WHAT CHANNEL PARTNERS THINK WHY NOT TO USE IoT?'

SCENARIO 1: Employers may use personal monitoring against us.

Research shows that almost 75 percent of employees believe their employer is capturing data about them without their knowledge, [67] so people are worried. In every industry, if employees feel that a new technology or management system provides no real benefit to them, they may avoid using it or even actively undercut its adoption.

What next: how to win the trust is the biggest challenge. Rethinking has to be done to design workplace IoT applications to offer employees obvious, and tangible benefit even though developers' ultimate aim may be to improve the supply chain performance.

SCENARIO 2: Channel Partners hesitant to integrate IoT in SCM.

Use of IoT generate huge volume of real data which is complex and heterogeneous in nature. When everything around in supply chain is quantified in number, the evaluation is based on not only performance metric and improvement but also to increase the level of improvement.

What next: Supply chain is more about managing relationships and people rather than revolving and dealing with big data. The challenge is to think how to navigate and explore the new opportunities with these.

One of the biggest challenge in the IoT based supply chain is the way IoT is perceived by channel partners. Top level of supply chain plays an important role in handling these perception and motivates everyone to contribute in the usage of IoT which can reap enormous benefits to the supply chain.

Introducing a 3 layered framework (See Figure 2) which depicts the interconnections between HR Management, SCM and IoT. The lowest layer, i.e. IoT is having 3 layers architecture – Sensing, Network and Application layer are tightly coupled with supply chain. The next layer is supply chain which consists of Suppliers (Inbound), Manufacturer and Customer (Outbound). The topmost layer being HR Management comprises of IT Governance structure, people, process of handling IoT, and organizational culture.

Let's look into the different components of HR Management. IT Governance is defined as 'establishing chains of responsibility, authority, and communication to empower people.' The key concepts is defined as 'RACE':

- Responsibility – being held accountable for a specific duty, task, or decision
- Authority – the power to influence behavior
- Communication – exchanging information

- Empowering – giving official authority to act

'RACE' has to be looked at individual as well as at the channel level.

- People – The people include the suppliers, manufacturer, transporter, wholesaler, distributor, dealer, retailer, customer, outsourcer, etc.
- The process is the outline of how to get to the destination. This include policies, procedures and task. The policy is the list of rules or the framework or the guidelines for the task. In this case, the policy is the rules and regulations for using IoT devices at various level in SCM. The procedure is the list of exact instructions that people needs to take to get the benefit as expected.
- Organization Culture - This is understood as the shared values of a group that become visible in actions and structures [68], [69].

The Figure 2 (see Appendix) explains the association between HR management and SCM with IoT. This overall integration poses great challenges on HR management as they have to keep their skills updated continuously. Acceptance and usage of IoT is dependent on employee skills which allows companies to better deal with supply chain integration. Channel partners attitudes and expectation from IoT significantly contribute in changing the face of the supply chain. This also impacts the performance and behaviour of employees both at individual and at channel level. As the supply chain have turned more complex, workforce would be in need to utilize the specialized devices and they need to interface viably over an assortment of supply chain. This is clear from the above chart that a supply chain capability to surpass in such environment will require a solid responsibility from HR Management.

Objective 4: To frame the best practices for motivating individuals and the channel partners to use IoT in supply chain.

The perception as discussed above present a significant challenge to channel partners when dealing with integrating SCM with IoT. They should have a sound strategy for maximum agility when absorbing the impact of IoT in supply chain. This primarily refers to having the right, skilled and motivated workforce composition to effectively use IoT as and when required.

It all comes down to educating and preparing the end to end supply chain to do the effective management by handling the concerns of channel partners. They must focus on people, and process to minimize vulnerabilities in this journey.

Following are roles and responsibilities that the channel partners must address -

- Setting up clear governance structure is considered the topmost priority when it comes to process accountabilities. Particularly, the specification of different roles of channel partners, such as sponsor, owner, and expert with their respective responsibilities are must.
- Motivating continuously through best scenarios and similar examples can be a driver of individual commitment for attainment of supply chain objectives.
- Monitoring and conducting chat sessions, bulletin boards, and knowledge sharing forums, addressing to

web portals, video conferencing for addressing issues and concerns raised by numerous channel partners in the chain is preferred.

- Providing the training to handle IoT and getting skilled staff, labor, and employees at all levels in the supply chain.
- Conducting training and awareness programs on periodic basis for better performance in supply chain.
- Train the training model could be adopted.
- Need to involve all stakeholders of the supply chain.
- Effective and proper communication ‘How to handle IoT’ so that real time information can be shared with all in supply chain, by keeping the privacy of the data as well as people.
- Need to tie up with Information Security Professional and their team as they are responsible for
 - Keeping the security checks and control in accessing the data.
 - Assessing trade-offs between security/confidentiality and access controls for the usage of IoT.
 - Need to inform about benefits and value added services of IoT in SCM by keeping checks on the negative factors.
 - Keep the checks on internal and external auditing in the process of handling IoT.
 - Keep updating the guidelines and procedures checks for using IoT.

Based on the review of the literature and in-depth analysis the best practices are framed and suggested for channel partners for using IoT in supply chain (Table 1).

An IoT-enabled supply chain encompasses a total lifecycle view of the product. IoT adoption and implementation would certainly benefit both individual and supply chain at large. “IoT is set to have a major impact on the manufacturing sector and it’s encouraging that supply chain and businesses expect potential benefits that technology can bring to provide better insights, proactive actions and improved outcomes,” said Nayaki Nayyar, GM & head of IoT [70]. IoT can enable end-to-end transformation for manufacturing companies and connect the shop floor to the top floor, optimizing end to end supply chains and backward and vertical integration and ultimately helping them stay competitive.

Table 1: Best practices for channel partners to use IoT in SCM

Individual Expectations	Always being connected has become need, habit and an expectation, so need to be very careful what to share and at what level. People are habituated to the “I accept” button and warning messages – don’t read what they are accepting or think about the consequences of this. Avoid such kind of situations. “Taking the easy way out always wins over security”, never choose the alternative ways to access the information. The desire to be connected, to download
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	applications, to share information with people online may invite hackers to access unauthorized data / information. Incentives for insecure behavior mean that security risks are ignored, such cases has to be reported immediately. Individual motivational level should be high to handle the situation about what to do and how to do it, and hence skill set has to be there to detect fraudulent activity at your own level. Avoid any situation which indicate a sign of trust/intimacy to share information including passwords and devices.
Channel Partners Expectations	Must have the Technology Governance framework having the details of roles and responsibility of IoT usage in SCM. Accountability plays an important role. Auditing mechanism should be in place. Need to identify all types of risks while using IoT in SCM. Have a reasonable and consistency security strategy set up, including portable BYOD approach. Backup their information, so that a digital assault does not wipe them out of their scholarly resources. Have an alternate course of action set up to know how precisely to go about when a digital assault strikes. The arrangement for how to proceed with business opportunity from a substitute area. Educate workers on best practices, how to distinguish phishing endeavors, safe versatile practices and so forth. Conduct representative individual verifications completely to avoid odds of insider dangers. A security concurrence with the representatives may likewise be in order.

IoT will promote new partnerships connected around new kinds of devices and integrating technology and channels of distribution. The overall findings are summarized in Table 2.

Table 2: Summarizing the findings and results

Objective 1	The value an organization creates with its supply chain has been determined largely by how well it manages links with their channel partners by keeping the expectations of their customers up to the mark. In the process, IoT plays a critical role. To integrate IoT with supply chain, the study reveals that the prerequisite is to understand a 3 layered architecture of IoT, which is a complex task. This integration is further used to enhance the functionality associated in SCM and the end to end transformation.
Objective 2	IoT is beneficial not only in supply chain but also across various other industries globally. The researcher has listed the positives and negatives influencing factors to integrate IoT in supply chain. Though it offers lot of positives but it rises the concern for the privacy of individuals or groups of people too. The major concerns are privacy, security and the presence of global universal standards. Though technology advancements could handle few of them, like mitigating the risk of handling security. The current state of interoperability standards remains a major concern.
Objective 3	After understanding how beneficial IoT is and it can enable end to end transformation by connecting to all its channel partners in real world, the researcher addressed that handling perceptions of people to use or not to use IoT is crucial in supply chain. Winning the trust of the channel partners to use IoT in supply chain is an important and challenging task. The researcher also proposed a framework which explains the association between HR management and SCM with IoT.
Objective 4	The findings identifies the best practices for motivating people and channel partners for using IoT in supply chain. Communication alone won't serve the purpose, right governance framework and addressing the concerns raised by people to handle IoT and getting skilled staff, labor, and employees at all levels in the supply chain is the need of an hour.

5. CONCLUSION

IoT is said to be disruptive and important when it comes to inter linked with supply chain strategy. One of the survey conducted by SCM World where the participation is from more than thousand supply chain executives, found IoT comes out to be most important technologies in the industry Since 2014. According to latest figures from the UK Government [71] there are 3.6 million registered businesses in the UK alone, that's nearly 1.8 million that are considering IoT adoption to improve their business performance in the near-term. Another finding conducted by Forrester Consulting on behalf of Zebra Technologies says that, "organizations recognize IoT as a strategic initiative for their

supply chain and they can achieve a wide range of business benefits from these solutions, especially improved customer experience."

The rise of enthusiasm for supply chain management has expanded as of late. But, most usage have not been founded on mechanized and automated supply chain management yet. Consolidation of the existing capabilities of SCM with current advancements like use of IoT will ensure the effectiveness in supply chain. This will deliver the computerized supply chain frameworks required in a dynamic environment. Supply chains can outfit the force of current advancements to drastically enhance the supply chain network execution by embracing such new points of view. This paper looked into the cutting edge innovative work of information sharing for proficient supply chain management. Advancements will demonstrate that they take into consideration collaborative SCM, Supply chain integration and supply chain management. As the essential empowering influence for compelling supply chain has been information sharing and will be upheld by advances in data innovation. Nonetheless, there are still numerous difficulties and issues that should be determined for SCM frameworks to end up being more material.

There is a trust that effective SCM depends more upon HR Management. With better HR Management, the supply network will move forward. At last, yet all the more significantly, this paper addresses how SCM is more than what can be executed with existing innovations; it demonstrates how positive and spurred HR Management can shape end to end supply network which is integrated with IoT. It is seen very clearly that technology innovation are driving to deal with the supply chain issues from both the tactical and strategic methodology [72]. Additionally it is well trusted that integrated SCM with IoT can be proficient with proper integration of the HR Management concerns tended to at individual and channel level. This paper will help researchers and practitioners to identify the contributing variables for improving supply chain performance and propose a structure on future SCM framework.

6. MANAGERIAL IMPLICATIONS OF THE RESEARCH

It is expected that finding of this paper will be extremely helpful and can be applied for both academicians and professionals. It would not only expand the degree of learning at the scholastic level, but may also help practitioners to fathom the importance and role of HR Management in picking up SCM adequacy by IoT. This will help the managers to achieve channel fulfilment and satisfaction which is extreme and crucial objective of every supply chain.

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8. APPENDIX

Figure 1 and Figure 2 are shown as below:

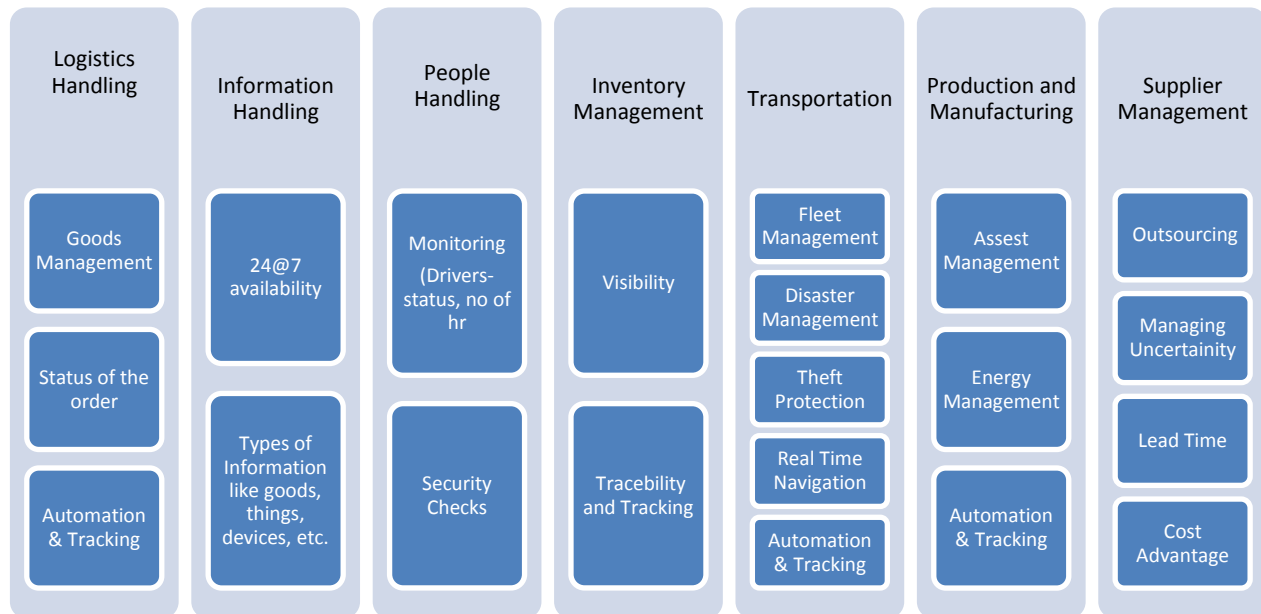


Figure 1: Functionality of SCM handled by IoT with their respective benefits

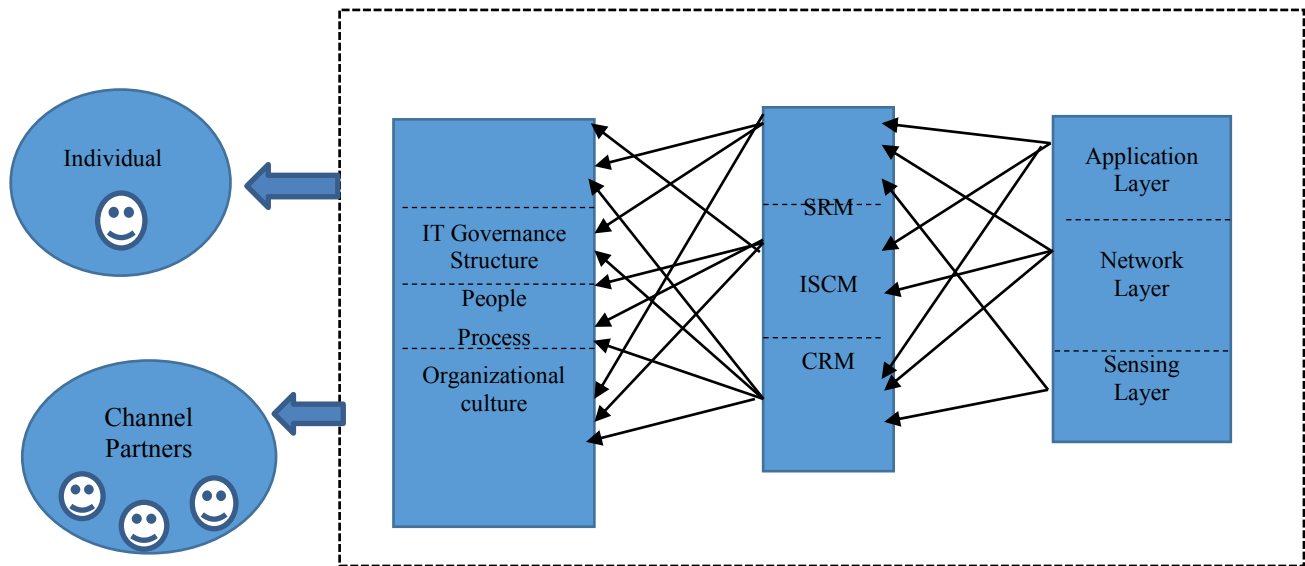


Figure 2: Proposed Three layer Framework to integrate HR Management with SCM and IoT